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Questions

Q1.

Which of these reacts directly with ethanoic acid to form ethanoyl chloride?

- ☐ **A** chlorine
- ☐ **B** chloroethane
- ☐ **C** hydrogen chloride
- ☐ **D** phosphorus(V) chloride

(Total for question = 1 mark)

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Q2.

The compound $HOOCCH{=}CHCOOH$ reacts with excess sodium hydroxide solution.

What is the organic product formed in the reaction?

- ☐ **A** $NaOOCCH{=}CHCOONa$
- ☐ **B** $HOOCCH{=}CHCOONa$
- ☐ **C** $NaOOCC(OH)HC(OH)HCOONa$
- ☐ **D** $NaOOCCH{=}CHCHO$

(Total for question = 1 mark)

Q3.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An organic compound that reacts with **both** lithium tetrahydridoaluminate(III) (lithium aluminium hydride) **and** magnesium could be

- ☐ **A** an aldehyde
- ☐ **B** a carboxylic acid
- ☐ **C** a ketone
- ☐ **D** a tertiary alcohol

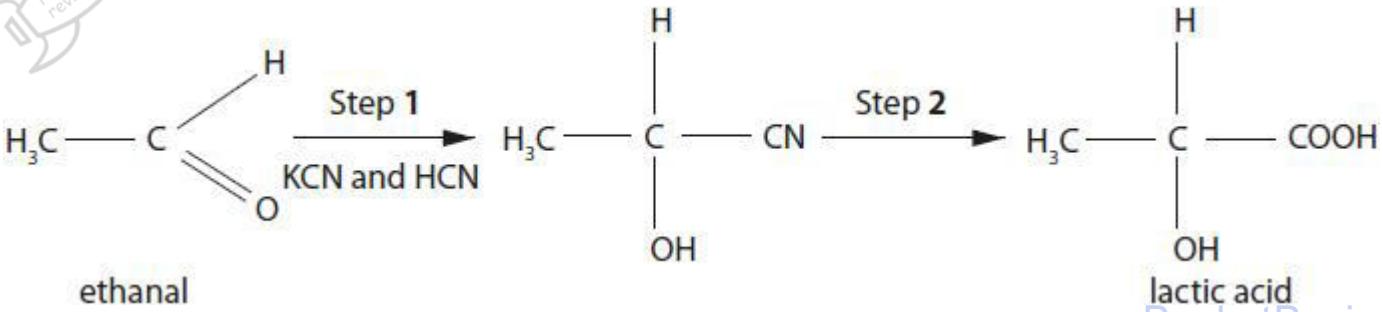
(Total for question = 1 mark)

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Q4.

The compound lactic acid can be synthesised from ethanal in two steps.



ethanal lactic acid

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(a) (i) Give the mechanism for Step **1**. Include curly arrows, and any relevant lone pairs and dipoles.

(4)

(ii) A student predicted that the product of Step **1** would rotate the plane of plane-polarised light. Comment on this prediction.

(3)

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(iii) Complete the table that summarises information about Step **2**.
State symbols are not required for the equation.

(4)

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Conversion of $CH_3CH(OH)CN$ to lactic acid	
Reaction type	
Reagent	
Conditions	
Equation	

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(b) Sodium hydrogencarbonate, $NaHCO_3$, has been used by some athletes to help prevent lactic acid causing muscle pain during exercise.

Write an equation for the reaction between sodium hydrogencarbonate and lactic acid.

(1)

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(c) Sodium hydrogencarbonate is part of a buffer in the body that controls the pH of blood. Two of the equilibria involved in this process are shown.

Equilibrium 1 $HCO_3^- + H_3O^+ \rightleftharpoons H_2CO_3 + H_2O$

Equilibrium 2 $H_2CO_3 \rightleftharpoons CO_2 + H_2O$

(i) Use the equilibria to explain how the buffer keeps the pH of blood nearly constant when a small increase in the concentration of hydrogen ions occurs.

(3)

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(ii) The pH of a blood sample was found to be 7.41.

Calculate the ratio of the concentration of HCO_3^- to H_2CO_3 in the blood sample.

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$$H_2CO_3 + H_2O \rightleftharpoons HCO_3^- + H_3O^+$$

$$K_a = 4.50 \times 10^{-7} \text{ mol dm}^{-3}$$

(3)

(Total for question = 18 marks)

Q5.

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This question is about lactic acid, $CH_3CH(OH)COOH$.

(a) Give the IUPAC name of lactic acid.

(1)

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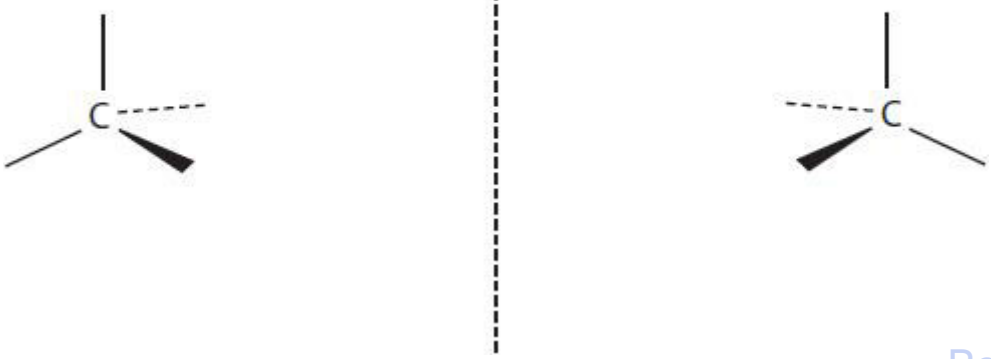
(b) Lactic acid is a chiral molecule, existing as a pair of non-superimposable mirror images called enantiomers.

Complete the 3D diagrams of the enantiomers of lactic acid.

(2)

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(c) Racemic lactic acid is synthesised in a two-step process.

In Step **1**, ethanal, CH_3CHO , is reacted with hydrogen cyanide, HCN , in the presence of KCN to form compound **X**.
In Step **2**, compound **X** is hydrolysed using dilute acid.

$$CH_3CHO + HCN \xrightarrow{\text{Step 1}} \text{compound X} \xrightarrow{\text{Step 2}} CH_3CH(OH)COOH$$

(i) State what is meant by **racemic** lactic acid.

(1)

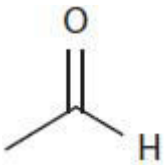
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(ii) Complete the mechanism for the formation of **X** in Step **1**.
Include curly arrows, and any relevant lone pairs and dipoles.

(4)



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(iii) Write the **ionic** equation for the hydrolysis of **X** in Step **2**.
State symbols are **not** required.

(2)

(iv) Explain, with reference to your mechanism in (c)(ii), why the lactic acid formed in this process is racemic.

(2)

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(d) Lactic acid is a weak acid.

(i) Calculate the mass, in g, of lactic acid, HA, required to make 1.00 dm^3 of an aqueous solution with a pH of 3.00.
Assume:

- $[HA]_{initial} = [HA]_{equilibrium}$
- $[H^+] = [A^-]$

[Lactic acid data: $pK_a = 3.86$ $M_r = 90.0$]

(4)

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(ii) In practice, the initial concentration of lactic acid does not equal its equilibrium concentration.
Explain how this affects the mass of acid needed to make a solution with a pH of 3.00. No calculation is required.

(2)

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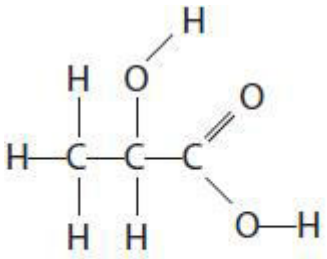
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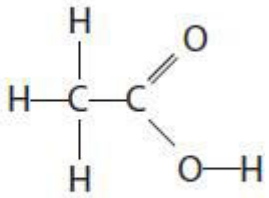
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(iii) Lactic acid is a stronger acid than ethanoic acid.



lactic acid



ethanoic acid

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Give a possible reason why lactic acid is the stronger acid.

(1)

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(Total for question = 19 marks)

Q6.

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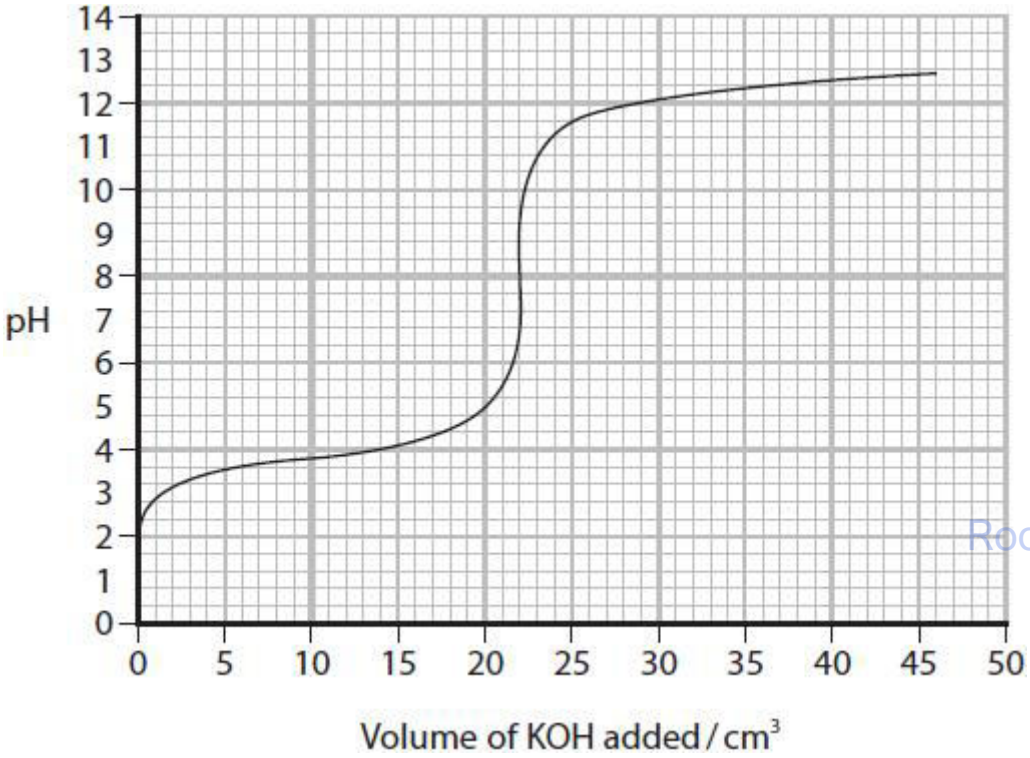
This question is about methanoic acid and propanoic acid.

(a) A student carried out a titration to find the concentration of an aqueous solution of potassium hydroxide.

25.00 cm^3 of 0.150 mol dm^{-3} aqueous methanoic acid was pipetted into a conical flask. Potassium hydroxide was added from a burette while measuring the pH. The titration curve is shown.

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(i) Complete the equation for the reaction taking place in this titration. State symbols are not required.

(1)

HCOOH +

(ii) Calculate the concentration of the potassium hydroxide solution, using your equation in (i) and the titration curve.

(2)

(iii) Use the titration curve to determine a value for the acid dissociation constant, K_a, of methanoic acid.

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(3)

(b) A student prepared a buffer by mixing together equimolar aqueous solutions of propanoic acid, CH_3CH_2COOH, and sodium propanoate, CH_3CH_2COONa.

The acid dissociation constant, K_a, for propanoic acid is 1.3×10^{-5} mol dm^{-3}.
Calculate the volume ratio of propanoic acid to sodium propanoate needed to produce a buffer

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solution
with a pH of 4.6

(2)

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(Total for question = 8 marks)

Q7.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Compound **X**, $C_4H_{10}O$, is oxidised to form compound **Y**.

Compound **Y** reacts with ethanol in the presence of concentrated sulfuric acid to give an ester, **Z**.

Which of these could be the formula of **Z**?

- ☐ **A** $CH_3COOC(CH_3)_3$
- ☐ **B** $CH_3CH_2COO(CH_2)_3CH_3$
- ☐ **C** $(CH_3)_2CHCOOCH_2CH_3$
- ☐ **D** $CH_3CH_2COOCH_2CH(CH_3)_2$

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(Total for question = 1 mark)

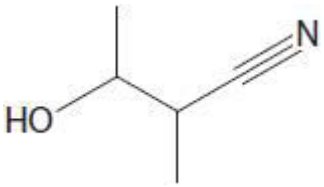
Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which acid would be produced by the hydrolysis of the molecule shown?

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- ☒ **A** 2-hydroxy-3-methylbutanoic acid
- ☒ **B** 3-hydroxy-2-methylbutanoic acid
- ☒ **C** 3-hydroxy-2,3-dimethylpropanoic acid
- ☒ **D** 4-hydroxypentanoic acid

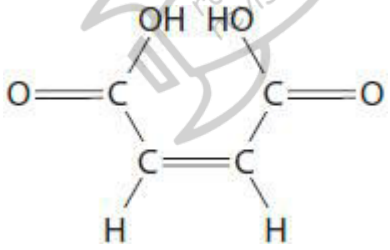
(Total for question = 1 mark)

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Q9.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The structure of maleic acid is shown.



(a) Which could be the titration curve when sodium hydroxide is added to maleic acid?

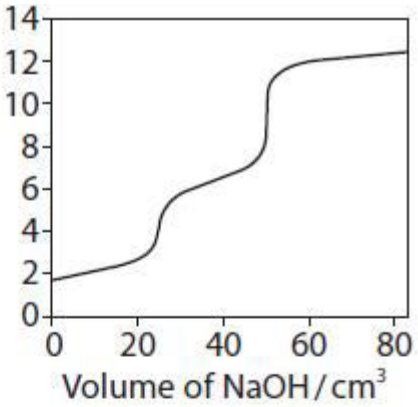
(1)

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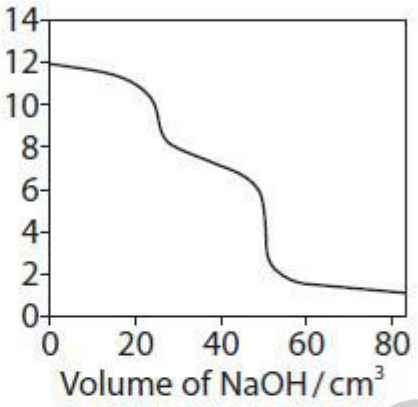
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☐ **A**

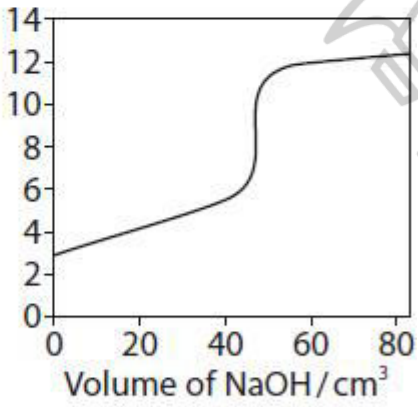


☐ **B**

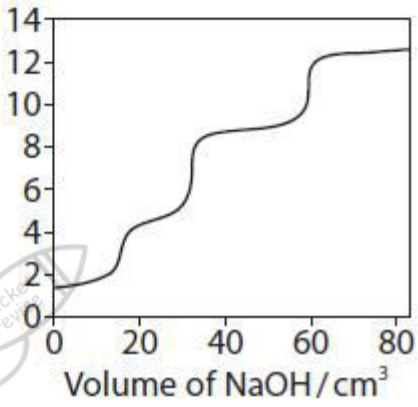


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☐ **C**



☐ **D**



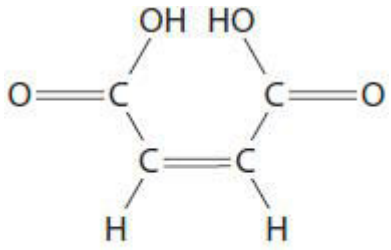
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(b) What is the IUPAC name for maleic acid?

(1)

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maleic acid

- ☒ **A** (*E*)-but-2-enedioic acid
- ☒ **B** (*Z*)-but-2-enedioic acid
- ☒ **C** (*E*)-1,2-ethenedioic acid
- ☒ **D** (*Z*)-1,2-ethenedioic acid

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(Total for question = 2 marks)

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

How do the boiling temperature and the solubility in water of butanoic acid compare with the values for hexane?

	Boiling temperature	Solubility in water
☒ **A**	lower	lower
☒ **B**	lower	higher
☒ **C**	higher	lower
☒ **D**	higher	higher

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(Total for question = 1 mark)

Q11.

Which compound has the highest boiling temperature?

(1)

- ☒ **A** butanal
- ☒ **B** butan-1-ol
- ☒ **C** butanone
- ☒ **D** butanoic acid

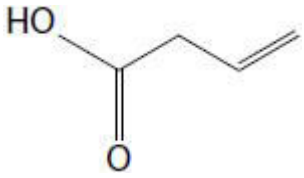
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(Total for question = 1 mark)

Q12.

Lithium tetrahydridoaluminate(III), $LiAlH_4$, is used to reduce the compound shown.



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The product formed is:

(1)

- ☒ **A** a saturated alcohol
- ☒ **B** a saturated carboxylic acid
- ☒ **C** an unsaturated alcohol
- ☒ **D** an unsaturated ketone

(Total for question = 1 mark)

Q13.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

But-2-ene-1,4-diol may be converted into 2-oxobutanedioic acid in a two-step synthesis through an intermediate compound **W**.

Identify the intermediate compound **W** and the reagent for Step **2**.

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	Compound W	Reagent for Step 2
☒ **A**	$HOCH_2CH(OH)CH_2CH_2OH$	$LiAlH_4$ in dry ether
☒ **B**	$HOOCH_2CH{=}CHCOOH$	$LiAlH_4$ in dry ether
☒ **C**	$OHCCH(OH)CH_2CHO$	hot acidified $K_2Cr_2O_7$
☒ **D**	$HOCH_2CH(OH)CH_2CH_2OH$	hot acidified $K_2Cr_2O_7$

(Total for question = 1 mark)

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Q14.

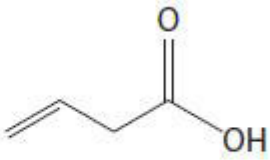
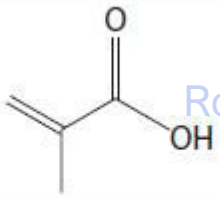
This question is about carboxylic acids and their derivatives.

(a) There are five carboxylic acids with the formula $C_4H_6O_2$.

Four are structural isomers.
One of these structural isomers has two stereoisomers.

(i) Name the two structural isomers shown.

(2)

Isomer		
Name		

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(ii) The third structural isomer has two stereoisomers.
Draw these stereoisomers.

(2)

Stereoisomer **1**	Stereoisomer **2**

(iii) Draw the fourth structural isomer.

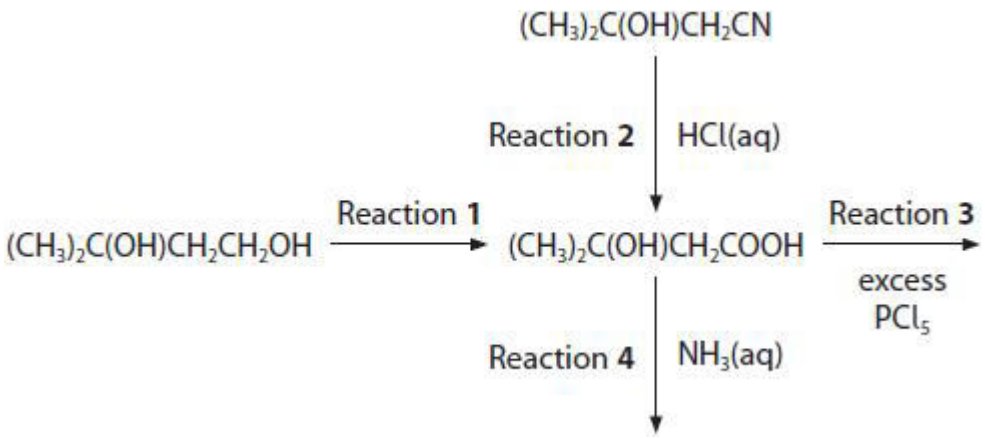
(1)

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(b) Some reactions involving 3-hydroxy-3-methylbutanoic acid, $(CH_3)_2C(OH)CH_2COOH$, are shown.

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(i) Give the reagents and conditions for Reaction **1**.

(1)

.....

.....

(ii) Write the equation for Reaction **2**. State symbols are **not** required.

(2)

(iii) Give the structure of the organic product formed in Reaction **3**.

(2)

(iv) Give the structure of the organic product formed in Reaction **4**.

(1)

(Total for question = 11 marks)

Q15.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which compound will have the **highest** boiling temperature?

- ☐ **A** $CH_3CH_2CH_2CH_2CH_3$
- ☐ **B** $CH_3CH_2CH_2CHO$
- ☐ **C** $CH_3COCH_2CH_3$

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☒ **D** CH_3CH_2COOH

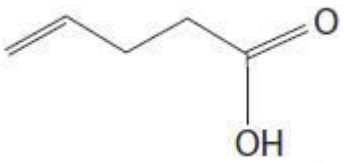
(Total for question = 1 mark)

Q16.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

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The compound shown reacts with **excess** $LiAlH_4$.



(a) What solvent is used in this reaction?

(1)

☒ **A** ethanol
☒ **B** ether
☒ **C** ethyl ethanoate
☒ **D** water

(b) What is the organic product that forms?

(1)

☒ **A**


☒ **B**


☒ **C**


☒ **D**


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(Total for question = 2 marks)

Q17.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Which substance is the least soluble in water?

- ☒ **A** propanal
- ☒ **B** propan-1-ol
- ☒ **C** propanoic acid
- ☒ **D** sodium propanoate

(Total for question = 1 mark)

Q18.

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Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which test can be used to distinguish between each of the pairs of compounds shown?

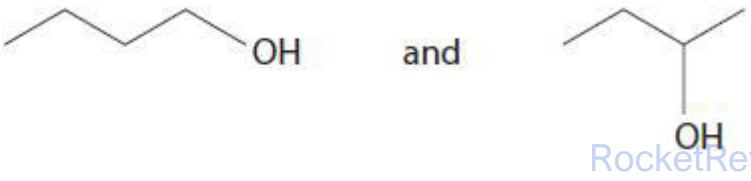
(a)



(1)

- ☒ **A** warm each compound with Tollens' reagent
- ☒ **B** add sodium carbonate solution to each compound
- ☒ **C** warm each compound with iodine under alkaline conditions
- ☒ **D** add phosphorus(V) chloride to each compound

(b)

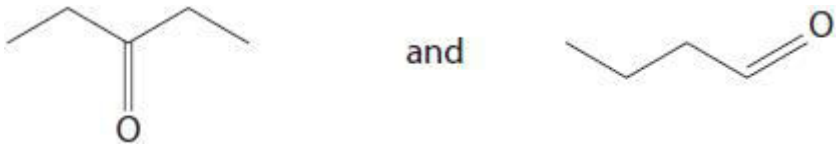


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(1)

- ☒ **A** warm each compound with Tollens' reagent
- ☒ **B** add sodium carbonate solution to each compound
- ☒ **C** warm each compound with iodine under alkaline conditions
- ☒ **D** add phosphorus(V) chloride to each compound

(c)



(1)

- ☒ **A** warm each compound with Tollens' reagent

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- ☒ **B** add sodium carbonate solution to each compound
- ☒ **C** warm each compound with iodine under alkaline conditions
- ☒ **D** add phosphorus(V) chloride to each compound

(Total for question = 3 marks)

Q19.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The reaction between ethanoic acid and lithium tetrahydridoaluminate(III) is best described as

- ☒ **A** elimination
- ☒ **B** oxidation
- ☒ **C** reduction
- ☒ **D** substitution

(Total for question = 1 mark)

Q20.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which reducing reagent is used to convert carboxylic acids to alcohols?

- ☒ **A** acidified potassium dichromate(VI)
- ☒ **B** concentrated sulfuric acid
- ☒ **C** hydrogen gas with a nickel catalyst
- ☒ **D** lithium tetrahydridoaluminate(III) in dry ether

(Total for question = 1 mark)

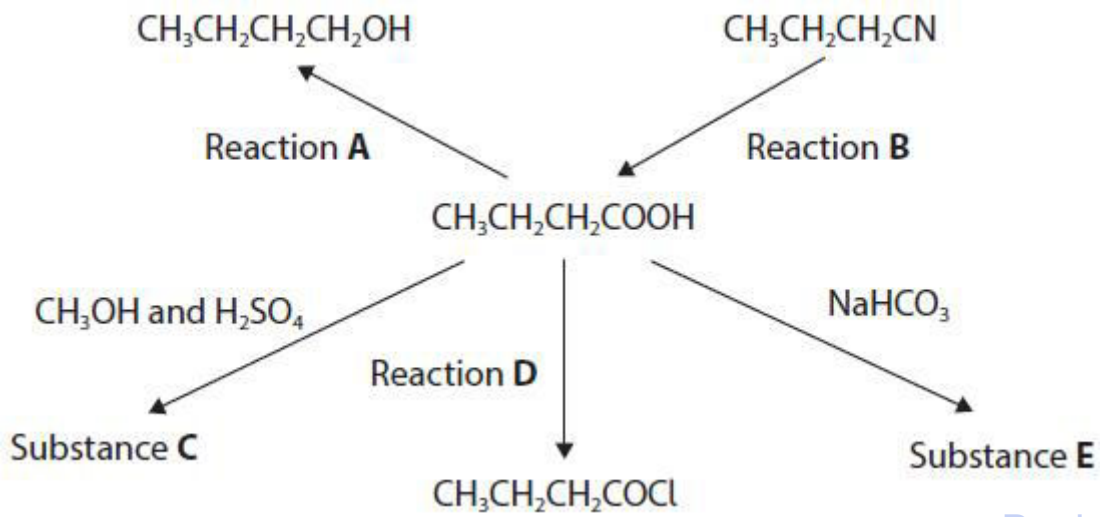
Q21.

This question is about butanoic acid.

(a) Some reactions involving butanoic acid are shown.

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(i) Identify the reagent, by name or formula, **and** the conditions required to carry out Reaction **A**.

(2)

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(ii) Name the type of reaction taking place in Reaction **B**.

(1)

.....

.....

(iii) Give the structural formula **and** name of Substance **C**.

(2)

.....

.....

(iv) Identify, by name or formula, the reagent required to carry out Reaction **D**, giving **one** observation.

(2)

Reagent .....

Observation .....

.....

.....

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(v) Write the full equation for the formation of Substance **E**.
State symbols are not required.

(1)

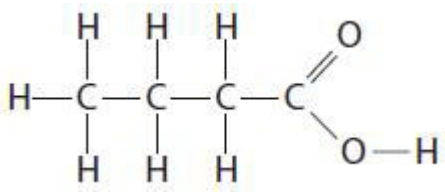
(b) Butanoic acid is very soluble in water. This is due to strong intermolecular forces between butanoic acid molecules and water molecules.

Complete the diagram by adding **one** water molecule to show this intermolecular force. Include relevant dipoles and lone pairs.
Label the intermolecular force involved.

(3)

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(Total for question = 11 marks)

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